

Scenario-Specific Paired Contrasts and Effect Sizes

M4 versus B1-B4 with confidence intervals and Cohen dz

Contrasts are calculated within each scenario using the ten matched seeds. The difference is defined as M4 minus baseline, so its sign must be interpreted using the direction of the endpoint.

Study	9 methods × 4 scenarios × 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Estimand and effect size

$$\delta_r = Y_{M4,r} - Y_{C,r}; \Delta = \text{mean}(\delta_r); d_z = \Delta / s_\delta$$

The 95% confidence interval is obtained from the ten paired differences. For mission time, $\Delta T < 0$ favors M4; for minimum distance, $\Delta_{\text{min}} > 0$ indicates greater M4 separation. Cohen d_z uses the standard deviation of paired differences rather than an independent-groups pooled standard deviation.

2. Results reproduced from the paper

Table 1. Each cell reports mean difference [95% CI] / d_z , where Δ = M4 - baseline.

Scenario	Comparator	ΔT [s] [95% CI] / d_z	Δ_{min} [m] [95% CI] / d_z
E1	B1	-2.97 [-3.40, -2.55] / -5.01	-0.240 [-0.406, -0.075] / -1.04
E1	B2	+0.92 [+0.16, +1.67] / +0.87	-0.057 [-0.139, +0.025] / -0.50
E1	B3	-0.51 [-0.70, -0.31] / -1.84	-0.015 [-0.073, +0.043] / -0.19
E1	B4	-1.90 [-2.04, -1.76] / -9.63	-0.409 [-0.494, -0.324] / -3.45
E2	B1	-3.34 [-3.54, -3.13] / -11.63	-0.253 [-0.375, -0.130] / -1.47
E2	B2	+1.95 [+1.78, +2.12] / +8.07	-0.131 [-0.172, -0.089] / -2.24
E2	B3	-0.33 [-0.58, -0.08] / -0.95	-0.133 [-0.156, -0.110] / -4.13
E2	B4	-2.07 [-2.22, -1.92] / -9.83	-0.107 [-0.184, -0.029] / -0.99
E3	B1	-3.24 [-3.46, -3.02] / -10.53	-0.233 [-0.396, -0.071] / -1.03
E3	B2	+2.26 [+2.01, +2.51] / +6.47	-0.131 [-0.173, -0.089] / -2.24
E3	B3	-0.08 [-0.33, +0.17] / -0.22	-0.128 [-0.173, -0.082] / -2.01
E3	B4	-2.01 [-2.27, -1.74] / -5.41	-0.339 [-0.391, -0.288] / -4.72
E4	B1	-2.27 [-2.61, -1.93] / -4.72	-0.130 [-0.217, -0.044] / -1.08
E4	B2	+2.79 [+2.35, +3.24] / +4.47	-0.130 [-0.175, -0.085] / -2.06
E4	B3	+0.51 [+0.17, +0.86] / +1.06	-0.115 [-0.157, -0.072] / -1.94
E4	B4	-0.81 [-1.16, -0.46] / -1.66	+0.209 [+0.171, +0.247] / +3.90

3. Reading the time contrasts

- M4 is faster than B1 and B4 in E1-E4; all corresponding intervals lie fully below zero.
- B2 is faster than M4 in all four scenarios; ΔT is positive from +0.92 to +2.79 s.
- M4 is faster than B3 in E1 and E2, similar in E3 because the CI crosses zero, and slower in E4.

4. Reading minimum-distance contrasts

- M4 has lower d_{min} than B1 in E1-E4.
- Against B2 and B3, E1 differences include zero; in E2-E4, M4 has lower d_{min} .
- Against B4, M4 has lower d_{min} in E1-E3 but higher d_{min} in E4 (+0.209 m).
- Greater d_{min} is not automatically better: the multicriteria analysis saturates safety utility at 1.20 m.

5. Multiplicity, Supplementary Table S1, and scope

The manuscript points to Supplementary Table S1 for the complete M4-versus-all-method set, all five endpoints, and raw and Holm-adjusted p values. This supplement reproduces Article Table 7 without inventing S1 values. Contrasts are scenario-specific and do not establish universal M4 superiority.